Mekong Quality Assurance Advisory

Keeping production true to what you approved

How the Product Specification, Packaging Specification and Verification Protocol fit together, and who owns each.

You approved a product. The risk in co-packing is that later production runs slowly stop matching it: a little darker, a little thinner, a different cap, a label with a printing error. Nobody changed the recipe on purpose, and nobody can prove what the standard was, because it was never written down tightly enough to check. I call that drift.

I prevent it with three working documents. The Product Specification fixes what your approved product is, as measurable ranges. The Packaging Specification fixes what the pack, label and pallet must be. The Verification Protocol is how I check each production run against both. Your counsel attaches them to the agreement with your co-packer, so they are enforceable.

I work from your approved finished product, not from your recipe. The recipe stays yours.

Naming. Your own paperwork may call these documents a schedule, annex or exhibit. What matters is that the content exists.

Scope: Quality and consistency verification against your specifications. Food safety stays with the plant.

The four documents at a glance

Document	What it is	Drift it prevents
Master Quality Agreement	The signed agreement between you and the co-packer that the other three documents attach to, written by your counsel: consequences of failure, change control, per-lot certificate of analysis, retained samples, traceability, recalls.	Nothing makes the plant answerable when a lot fails, and changes happen without sign-off.
Product Specification	The food: color, °Brix, pH, water activity, consistency, fill, visual, and taste and smell, as measurable ranges built from your approved product.	The product slowly moving away from the one you approved.
Packaging Specification	The pack: seal and closure, label and printing, ingredient list and allergen statement as printed, barcode, lot and date code, carton, pallet.	Quiet changes in materials and pallets, and printing errors that reach the shelf.
Verification Protocol	How each production run is checked against the other two: what I measure on site, what goes to a lab, sampling, defect classes, what happens when a lot fails.	The first two documents sitting on paper with nobody checking them.

The big picture

Teal: I write it or carry it out. Amber: you own or approve it. Gray: the plant or your regulatory advisor owns it.

MASTER QUALITY AGREEMENT *Owned by you and your counsel*

- Makes the three documents below binding. Holds consequences of failure (reject, retest, hold), change control, per-lot certificate of analysis, retained samples, traceability, recalls.

Drift prevented: nothing makes the plant answerable when a lot fails.

PRODUCT SPECIFICATION *I write it; you approve*

The food: quality and consistency (safety limits are not set here)

- Color, °Brix, pH and water activity: lab baselines, set as ranges
- Bostwick consistency, fill, visual (specks, gloss, separation)
- Taste and smell against a retained reference sample and a short flavor profile
- Named off-notes, zero tolerance: burnt, metallic, rancid, fermented
- Each attribute measured by a method defined for your product
- Initial ranges tighten by an agreed rule after the first lots

Built from your approved product, not your recipe.

Drift prevented: the product slowly moving away from the original.

PACKAGING SPECIFICATION *You own the standard; I record it*

The pack: bottle, closure, label, carton, pallet

- Path 1: your packaging consultant writes it
- Path 2: the plant states its specs, I record them, you approve
- Pack and seal, cap torque, cartons, pallet pattern and wrap
- Print quality and printing errors; barcode; lot and date code
- Ingredient list and allergen statement as printed, matched to your approved artwork (legal compliance is for your advisor)
- Light document check; not a structural engineering review

Drift prevented: quiet changes in materials, print and pallets.

VERIFICATION PROTOCOL *I write it and carry it out*

ON SITE, with my own instruments

- Bostwick consistometer, refractometer (°Brix), pH meter, water activity meter
- Color: per the product's defined method, after a calibration check
- Taste and smell against the reference; blind panel for quality calls
- Fill, seals, labels, print, codes; cartons, pallets, foreign matter
- AQL sampling (ANSI/ASQ Z1.4); critical, major and minor defects

SENT TO AN ACCREDITED LAB (about 3 to 4 days)

- Baseline readings when the specification is set: color, water activity, pH, °Brix
- Microbiology and other lab-only tests you ask for
- Disputed readings: retained split sample to a pre-agreed lab
- I collect, seal, ship and track every sample; the lab tests; results are reported against your specification
- Lot held until lab results are back
- One tamper-sealed sample kept from each inspection

Inspection report to you, with readings and photos

You decide: release, hold, rework or reject

FOOD SAFETY: owned by the plant and your regulatory advisor

- HACCP, process validation, certification, label legality. If I see red flags, I review key records and report and escalate.

Who owns what

Component	Prepared by	Approved by	My role	Where my role ends
Recipe and formulation	You	You	I work from the finished product you approve, not from the recipe.	Writing, changing or reviewing your recipe.
Master Quality Agreement	Your counsel	You and the co-packer	I supply the three documents to attach, and advise on what the agreement should require.	Drafting legal terms or giving legal advice.
Product Specification	Me	You (the co-packer acknowledges it)	I set the ranges from lab baseline readings, define the measurement method for each attribute, and set the taste, smell and appearance reference.	Setting food safety limits.
Packaging Specification	Path 1: your packaging consultant. Path 2: the co-packer states its specs.	You	Path 1: I turn it into inspection checks. Path 2: I request the plant's specs, record them, and run a light document check.	Engineering or designing the packaging, or certifying it will survive transit.
Verification Protocol	Me	You	I write it, carry out the inspections, and report the results.	Releasing or rejecting the lot. You decide.
Food safety and label legality	The co-packer; your regulatory advisor	Certification body or regulator, where applicable	If I see red flags, I review key records and report and escalate any safety-related finding.	Validating or certifying food safety, or ruling on whether a label is compliant.

Product Specification: what goes in it

The baseline comes from your approved product. For color, water activity, pH and °Brix, an accredited lab takes every baseline reading, using the method defined for that product, so the standard rests on traceable numbers. I set the consistency and sensory baselines myself. Each parameter has a target value and a range, and each is measured by a method defined for your product and written into the specification.

Initial ranges are time-boxed. For the first commercial lots (for example, the first three, or the first run if you produce less often), the ranges are initial ranges: they are outer limits and never loosen. After those lots they tighten by a method agreed in advance, such as recalculating from the first lots' data with a maximum width, and the tightened ranges take effect automatically, with no renegotiation. The target value stays fixed, and my reports track where each lot lands inside its range, so a plant drifting toward the loose edge is flagged even while lots still pass. Your counsel writes the automatic-tightening clause into the agreement.

The specification covers color, °Brix, pH, water activity, consistency, fill and visual, and taste and smell. Taste and smell is defined by a retained sealed reference sample and a short flavor profile written with you (aroma, main flavor notes, balance, texture and finish), plus a zero-tolerance list of named off-notes: burnt or scorched, metallic, rancid, fermented, bitter. How each attribute is checked, and with what, is in one table under the Verification Protocol.

What stays out of the Product Specification

Item	Where it lives	Owner
Allergens and cross-contact	The plant's food safety system. The allergen statement as printed is checked against your approved artwork through the Packaging Specification.	Co-packer; you for the label
Contaminants (heavy metals, mycotoxins, pesticide residues)	Your regulatory requirements. I can arrange lab testing if you ask.	Your regulatory advisor
Microbiology	The plant controls it. The Verification Protocol has a lab lane and a lot hold.	Co-packer (control); me (sampling and reporting)
Additives, colorants, destination-market labeling rules	Your regulatory compliance.	You
Traceability and lot coding	Lot code format in the Packaging Specification; traceability duties in the agreement.	You and the co-packer
Per-lot certificate of analysis, retained samples, change control, recalls, consequences	Master Quality Agreement.	Your counsel

Packaging Specification: two ways to build it

	Path 1: you have a packaging consultant	Path 2: you use the co-packer's packaging
Who defines the standard	Your consultant writes the Packaging Specification.	The plant states what it can do; you agree to it.
What I do	I turn the specification into inspection checks.	I request the plant's packaging specs (seal, cap torque, carton strength, pallet pattern, artwork, barcode), record them as the Packaging Specification, and run a light document check, not an engineering review: food-contact materials, whether supplier test data exists for the carton and pallet, and whether the shipping route calls for a packaging consultant or a transit test.
Your role	You approve the consultant's specification.	You approve the recorded specification before it becomes the standard.
What gets inspected	The consultant's standard.	What the plant says it can do and you agreed to.
What the record is	The consultant's engineering, which I inspect against.	A baseline description of what the plant uses. It is not a structural load assessment, and who carries the risk of packaging that fails in transit is decided in the agreement.
What I do not do	Engineer or redesign the packaging.	Engineer or redesign the packaging.

Contents: primary pack and seal; closure and cap torque; label and artwork, including print quality, smearing, misprints and color against the approved artwork; ingredient list and allergen statement as printed; net contents and claims as approved; barcode; lot and date code; master carton; pallet pattern and wrap.

I check that what is printed matches the artwork you approved. Whether that label is legally compliant is for your regulatory advisor.

In Path 2 the record describes what the plant uses. For a long or rough shipping route, I recommend a packaging consultant or a transit test, and your counsel should decide in the agreement who carries the risk if packaging fails in transit.

Verification Protocol: what I check and how

The lab takes the baseline readings once, when the specification is set. At each inspection, my on-site readings are checked against those baseline ranges. This one table shows each check, where its baseline comes from, and how it is done.

Check	Baseline	Where	Method	Notes
Color	Accredited lab readings on your approved product; ranges set from the results.	On site	Per the product's defined method, after a calibration check.	Color is locked in as a number, so it can't be argued after the fact. If the plant has no suitable instrument, a retained sample goes to the accredited lab.
°Brix	Accredited lab baseline readings.	On site	Refractometer.	
pH	Accredited lab baseline readings.	On site	pH meter.	A consistency indicator. The plant's own safety limit is recorded, not set, by me.
Water activity	Accredited lab baseline readings.	On site	Water activity meter.	Same treatment as pH.
Consistency	My baseline readings on your approved product.	On site	Bostwick consistometer.	Per the product's defined method.
Fill, seals, leaks and visual	Your declared fill and the approved product.	On site	Fill check; visual and handling checks for specks, gloss, separation, seals and leaks.	
Taste and smell	Retained sealed reference sample, a short flavor profile written with you, and named off-notes with zero tolerance.	On site	Side by side with the reference and profile. Quality calls by blind panel.	I smell first, and I never taste a sample that is swollen, leaking, or smells fermented or chemical. Spoilage and contamination off-notes are escalated, not put to a panel. If the panel can't agree, you decide.
Labels, printing, barcode, lot and date codes	Approved artwork in the Packaging Specification.	On site	Compared with the approved artwork; ingredient list and allergen statement read against it; barcode scanned.	Printing errors are recorded with photos.
Cartons and pallets	Packaging Specification.	On site	Against the Packaging Specification.	
Visible foreign matter		On site	Visual.	
Microbiology and other lab-only tests	Only where you ask.	Accredited lab	Results take about 3 to 4 days.	Lot held until results are back.

My instrument calibration records are kept and available to you, so my readings are defensible. I also check any instrument I don't own before I accept its readings.

If a reading is disputed

The first step is to measure the same sample side by side with the plant and compare calibration records. If it is still disputed, a retained split sample goes to a pre-agreed accredited lab, which measures it against the recorded numbers using the method named in the agreement. The binding terms, who pays for the lab, and the timing belong in the agreement your counsel writes.

Retained samples. The plant keeps sealed retains of each lot, and the number and the time they are kept are set in the agreement, scaled to the pack size. At each inspection I also take one tamper-sealed sample, log it and hold it, so a dispute sample never rests only on the plant's own storage.

How a taste or smell call is made

Each run is compared with the retained reference and the flavor profile written with you: aroma, main flavor notes, balance, texture and finish. A difference from the profile is reported. Off-notes are handled as below.

Spoilage and contamination off-notes (rancid, fermented, moldy, chemical or machine oil) are not put to a vote. I report and escalate immediately, hold the lot, and send a sample for lab analysis.

Quality calls (burnt or scorched, metallic, bitter, clearly off the reference) are made by a blind side-by-side comparison with the retained reference, by at least two tasters including the plant's QA person. A triangle test, where tasters pick the odd sample out of three, shows whether the production sample is reliably different from the reference. A quality hold never rests on my palate alone. If the panel can't agree, I give you the findings and the retained sample, and you decide.

Defect classes

Class	Examples	What happens
Critical	Visible foreign matter; leaks, seal failures or swollen packs; wrong product or wrong pack; missing or wrong allergen statement; spoilage or contamination off-notes (rancid, fermented, moldy, chemical or machine oil); any reading outside the plant's own validated safety limit.	I stop and report to you immediately. Lot on hold.
Major	°Brix, pH, water activity, color or consistency out of range; quality off-notes confirmed by the blind panel (burnt or scorched, metallic, bitter, clearly off the reference); fill out of range; printing errors or wrong claims on the label; wrong or unscannable barcode; unreadable lot or date code; crushed cartons; unstable pallet.	Counts against the AQL. A failed lot is held or rejected at your decision.
Minor	Slight label tilt; small cosmetic print flaws; smudged but legible date code; scuffed master cartons.	Counts against the AQL. Noted in the report.

Sampling uses AQL under ANSI/ASQ Z1.4, with limits set by lot size and product risk.

The food safety boundary

The plant controls food safety. This is who owns each part, and where I come in.

Item	Owner	Where I come in
Hazard analysis and process controls (HACCP)	Co-packer	I report and escalate anything safety-related that I see on site.
Process validation, including thermal process	Co-packer and its process authority	The validated process sets the limits I compare readings against.
Safety limits, such as a maximum pH or water activity	The plant's validated process, or your regulatory advisor	I record them as supplied, compare readings to them, and report and escalate anything outside.
Microbiological control	Co-packer	I collect and send samples to a lab if you ask, and report results against your specification.
Destination-market regulatory compliance and label legality	Your regulatory advisor	I check that print matches your approved artwork. Whether the label is legally compliant is for your advisor.
Records check, when I see red flags	Me (screening only)	I review key food safety records as a risk-lowering screen and report what I find. If concerns arise, I recommend a third-party food safety audit.

A records check is a screening step. It does not evaluate the plant's food safety system.

How an engagement runs

1. Baseline. You provide your approved product. An accredited lab takes baseline readings for color, water activity, pH and °Brix. I take the consistency baseline, write the flavor profile with you, retain the taste and smell reference, set the ranges, and define the measurement method for each attribute. This produces the Product Specification.

2. Packaging. Path 1 (your consultant writes it) or Path 2 (the plant states its specs and I record them). You approve. This produces the Packaging Specification.

3. Protocol. I write the Verification Protocol from the first two documents: instruments, on-site and lab lanes, AQL sampling, defect classes.

4. Agreement. You and your counsel attach the three documents to the Master Quality Agreement.

5. Inspect. I check each production run, scheduled or unannounced, and report with readings and photos. You decide to release, hold, rework or reject.

6. Review drift. After the first lots the ranges tighten by the method agreed in advance. I also update the documents through change control.